



ECONOMIC RESEARCH
FEDERAL RESERVE BANK OF ST. LOUIS
WORKING PAPER SERIES

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Authors	Alexander Bick, Adam Blandin, and David J. Deming
Working Paper Number	2024-027F
Revision Date	October 2025
Citable Link	https://doi.org/10.20955/wp.2024.027
Suggested Citation	Bick, A., Blandin, A., Deming, D.J., 2025; The Rapid Adoption of Generative AI, Federal Reserve Bank of St. Louis Working Paper 2024-027. URL https://doi.org/10.20955/wp.2024.027

Federal Reserve Bank of St. Louis, Research Division, P.O. Box 442, St. Louis, MO 63166

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The Rapid Adoption of Generative AI*

ALEXANDER BICK

Federal Reserve Bank of St. Louis & CEPR

ADAM BLANDIN

Vanderbilt University

DAVID J. DEMING

Harvard Kennedy School & NBER

October 27, 2025

Abstract

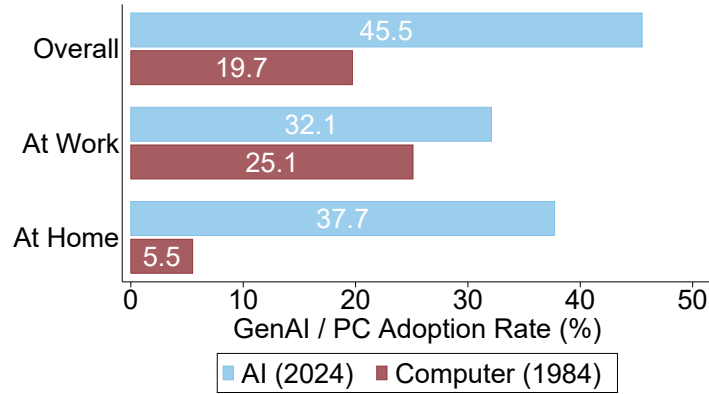
Generative artificial intelligence (AI) is a potentially important new technology, but its impact on the economy depends on the speed and intensity of adoption. This paper reports results from a series of nationally representative U.S. surveys of generative AI use at work and at home. As of late 2024, 45% of the U.S. population age 18-64 uses generative AI. Among employed respondents, 27% used generative AI for work at least once in the previous week: 10% used it every workday, and 17% on some but not all workdays. Relative to each technology's first mass-market product launch, work adoption of generative AI has been as fast as the personal computer (PC), and overall adoption has been faster than either PCs or the internet. Between 1 and 7% of all work hours are currently assisted by generative AI, and respondents report time savings equivalent to 1.4% of total work hours. Potential productivity gains vary widely by industry, and firm climate and policies play an important role in adoption patterns.

JEL Codes: J24, O33

Keywords: Generative AI, Technology Adoption, Labor Productivity

**Contact:* alexander.bick@stls.frb.org; adam.blandin@vanderbilt.edu; david_deming@harvard.edu. We thank Kevin Bloodworth II and Eli Tsung for excellent research assistance. The views in this paper are those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of St. Louis or the Federal Reserve System.

Figure 2: Computer and Generative AI Adoption 2-3 Years After First Mass-Market Product



Notes: The figure displays usage rates for computers (PCs) and genAI in 1984 and 2024, respectively. The IBM PC was released in 1981, which we refer to as the first mass-market computer, implying that 1984 is three years after the first mass-market computer. ChatGPT was released in November 2022, which we refer to as the first mass-market genAI product, implying the fall of 2024 is two years after the first mass-market product for that technology. The data source for genAI is the August and November 2024 wave of the RPS (blue bars). The data source for computers is the 1984 Computer and Internet Use Supplement of the CPS (red bars). The sample for the Overall and At Home bars in the RPS and CPS is all individuals ages 18-64 (RPS, $N = 9726$; CPS, $N = 61,708$). The sample for the For Work bars is employed individuals ages 18-64 (RPS, $N = 6935$; CPS, $N = 89,271$).

measured using very similar questions for both technologies.) The first mass-market computer was the IBM PC, which was released in August 1981. The first mass-market genAI product was ChatGPT, which was released in November 2022.⁸ For the PC, we measure adoption in the CPS CIU for the first time in 1984, i.e. three years after the first mass-market computer. The RPS provides adoption estimates for 2024, two years after the first mass-market genAI product.

The top pair of bars in Figure 2 refers to overall adoption for the two technologies two to three years after the introduction of the first mass-market product. The bar for the RPS repeats the 45% overall adoption rate for genAI reported in Figure 1a. Overall PC adoption is roughly 20% in 1984, implying that genAI has been adopted at a faster pace than the PC. As another more recent comparison, in 1995 the internet was opened to commercial traffic after the decommissioning of NSFNet (Leiner et al., 2009).⁹ Two years later, 21% of US adults reported access to the internet according to the International Telecommunication Union, which compiles subscriber data from national regulatory bodies and service providers (Pena-Lopez et al., 2009). This is a similar pace of overall adoption to PCs, and substantially slower than the adoption of genAI.

⁸Three notable computer products had already been released in 1977 (the Apple II, the Commodore PET, and the TRS-80). Choosing earlier introduction dates would imply slower adoption rates for a given technology. The 1981 IBM PC was the first computer to sell over one million units (Abbate, 1999). We chose these dates based in part on ChatGPT’s answers to questions about the year of the first mass market product for each technology. The exact prompts and answers are available upon request.

⁹1995 was also the year of Netscape’s initial public offering and the year that AOL 3.0 was released.

A key question is whether genAI’s faster adoption was driven by work or non-work use.¹⁰ The faster overall pace of genAI adoption compared with PCs is driven by non-work adoption. The middle pair of bars in Figure 2 plots work adoption of genAI and PCs. We find similar work adoption rates for genAI after two years (32%) and PCs after three years (25%). By contrast, the bottom pair of bars in Figure 2 reveals much higher non-work adoption rates for genAI (38%) compared with PCs (5%).

The computer and related information technologies induced far-reaching changes in firm organization and management, which led to substantial productivity gains (Brynjolfsson and Hitt, 1996). The similar pace of genAI adoption by workers thus far suggests that this new technology could also be transformative for firms. At the same time, the choice of whether to adopt a technology depends on both its benefits and costs. PCs required purchasing expensive immobile hardware and nontrivial training. In contrast, many genAI products are free or inexpensive and user-friendly. On the one hand, this lower adoption cost likely contributes to genAI’s rapid uptake. On the other hand, this means that similar adoption rates for PCs and genAI at work do not necessarily imply equivalent benefits.

4 Which Workers Use Generative AI?

In this section, we document that early genAI adoption has not been uniform throughout the workforce. Understanding which workers are adopting genAI at higher rates provides several important insights. First, because many workers are adopting genAI ahead of firm-level adoption, variation in genAI use can help managers allocate support, training, and integration efforts. Second, we show that employer encouragement of genAI use is highly predictive of adoption, suggesting that firms and managers can influence the nature and extent of genAI use in their organization. Third, adoption patterns shed light on how genAI maps onto specific tasks and roles within the firm, which is relevant for decisions regarding job redesign and firm reorganization.

Figure 3 documents demographic heterogeneity in work adoption of genAI. The top panel shows genAI adoption is 6 percentage points higher for men. The middle panel show genAI adoption by age. Workers aged 18-49 have similar adoption rates, while adoption is lower for workers aged 50-64. The bottom panel shows that adoption rates are increasing in educational attainment. In particular, workers with a Bachelor’s degree or more use genAI at roughly twice the rate of non-college educated workers. Differences between demographic groups are primarily driven by users who use genAI every workday or at least once in the previous week.

Figure 4 plots the kernel density of genAI adoption by weekly earnings percentile. Adoption

¹⁰Unfortunately, internet adoption data do not allow us to distinguish work from non-work adoption.